



## 4.2 Statistical measures and displays

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

### Understand the concept

Measures of central tendency (mean, median, mode) summarise a typical value; measures of spread (range, interquartile range  $IQR=Q3-Q1$ , variance, standard deviation  $\sigma$ ) describe variability. Standard deviation  $\sigma=\sqrt{(\sum(x-\bar{x})^2/n)}$  for population data. Box plots display the five-number summary (min,  $Q1$ , median,  $Q3$ , max); an outlier is typically a value below  $Q1-1.5\times IQR$  or above  $Q3+1.5\times IQR$ . Histograms display frequency distributions of grouped continuous data.

### Worked example: follow the reasoning

A data set is 4, 7, 8, 9, 12, 15, 20. Find the median,  $Q1$ ,  $Q3$ , and  $IQR$ .  
Median (4th value)=9.  
Lower half: 4,7,8  $\rightarrow Q1=7$ . Upper half: 12,15,20  $\rightarrow Q3=15$ .  
 $IQR=15-7=8$ .

### Quick check · multiple choice

A data set has mean 20 and standard deviation 4. A value of 32 is considered an outlier by which common rule?  
A It exceeds  $\text{mean}+2\sigma$   
B It exceeds  $\text{mean}+3\sigma$   
C It is above  $Q3+1.5IQR$   
D It is below the median

Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The following are the times (minutes) taken by 11 students to complete a task: [7]  
12, 15, 15, 18, 20, 21, 22, 24, 25, 30, 45.  
(a) Find the median,  $Q1$ , and  $Q3$ .  
(b) Calculate the  $IQR$  and determine whether 45 minutes is an outlier using the  $1.5\times IQR$  rule.  
(c) State, with a reason, whether the mean or the median is a more appropriate measure of central tendency for this data set.

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Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [7]

(a)  $n=11$ , median is 6th value=21. Lower half (5 values): 12, 15, 15, 18, 20  $\rightarrow Q1=15$  (3rd value). Upper half: 22, 24, 25, 30, 45  $\rightarrow Q3=25$  (3rd value).

(b)  $IQR=25-15=10$ . Upper boundary= $Q3+1.5(IQR)=25+15=40$ . Since  $45>40$ , it IS an outlier.

(c) The median is more appropriate, since the outlier (45) would pull the mean upward, making it unrepresentative of the typical student's time, whereas the median is resistant to outliers.

### Quick check answer

A.  $\text{mean}+2\sigma = 20+8=28$ , and  $32>28$ , so it exceeds two standard deviations above the mean (a common informal outlier rule, though the IQR rule is also standard for box plots).

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.